

## LAB MANUAL

### Ethical Hacking Laboratory – Kali Linux Commands and Nmap Scanning

Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

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### Experiment 21: Batch File Execution

**Aim:**

To execute batch files.

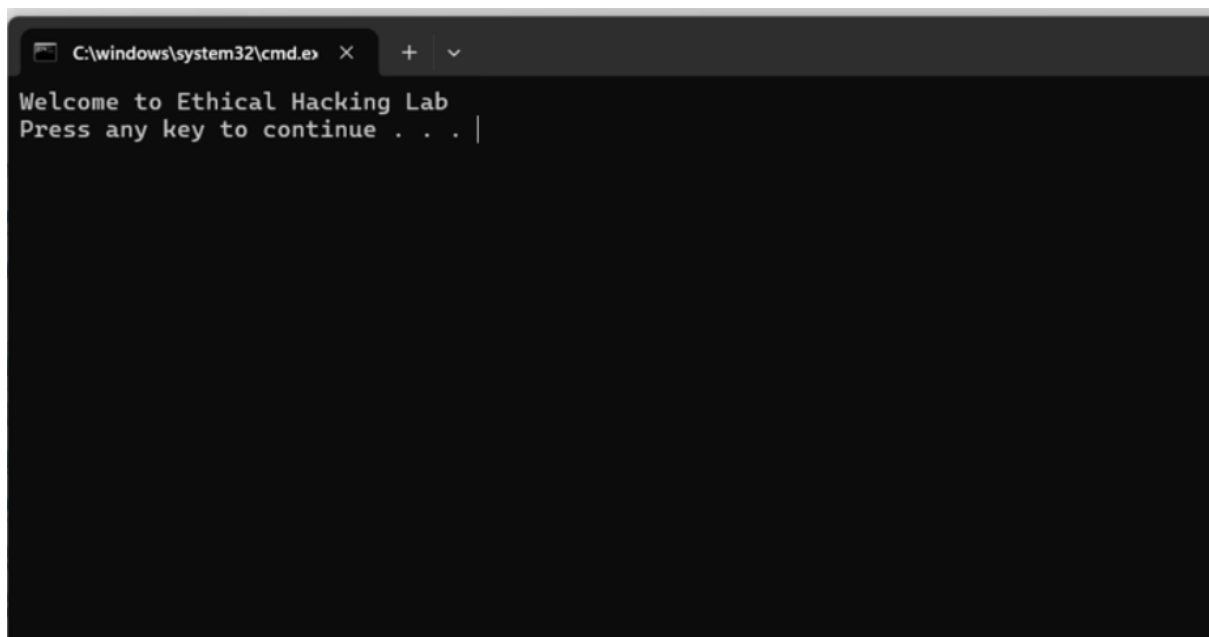
**Sample Batch File:**

```
@echo off
echo Welcome to Ethical Hacking Lab
pause
```

**Execution:**

sample.bat

**Output :**



```
C:\windows\system32\cmd.exe
Welcome to Ethical Hacking Lab
Press any key to continue . . . |
```

Successfully executed batch file.

## Experiment 22: Packet Analyzer Tool

**Aim:**

To analyze network packets.

## Tools:

- Wireshark
- Tcpdump

## Commands:

```
tcpdump -i eth0
```

**output**

[illegible]

**Result:**

Successfully analyzed packets.

## Experiment 23: Port Scanning Tools (Extended)

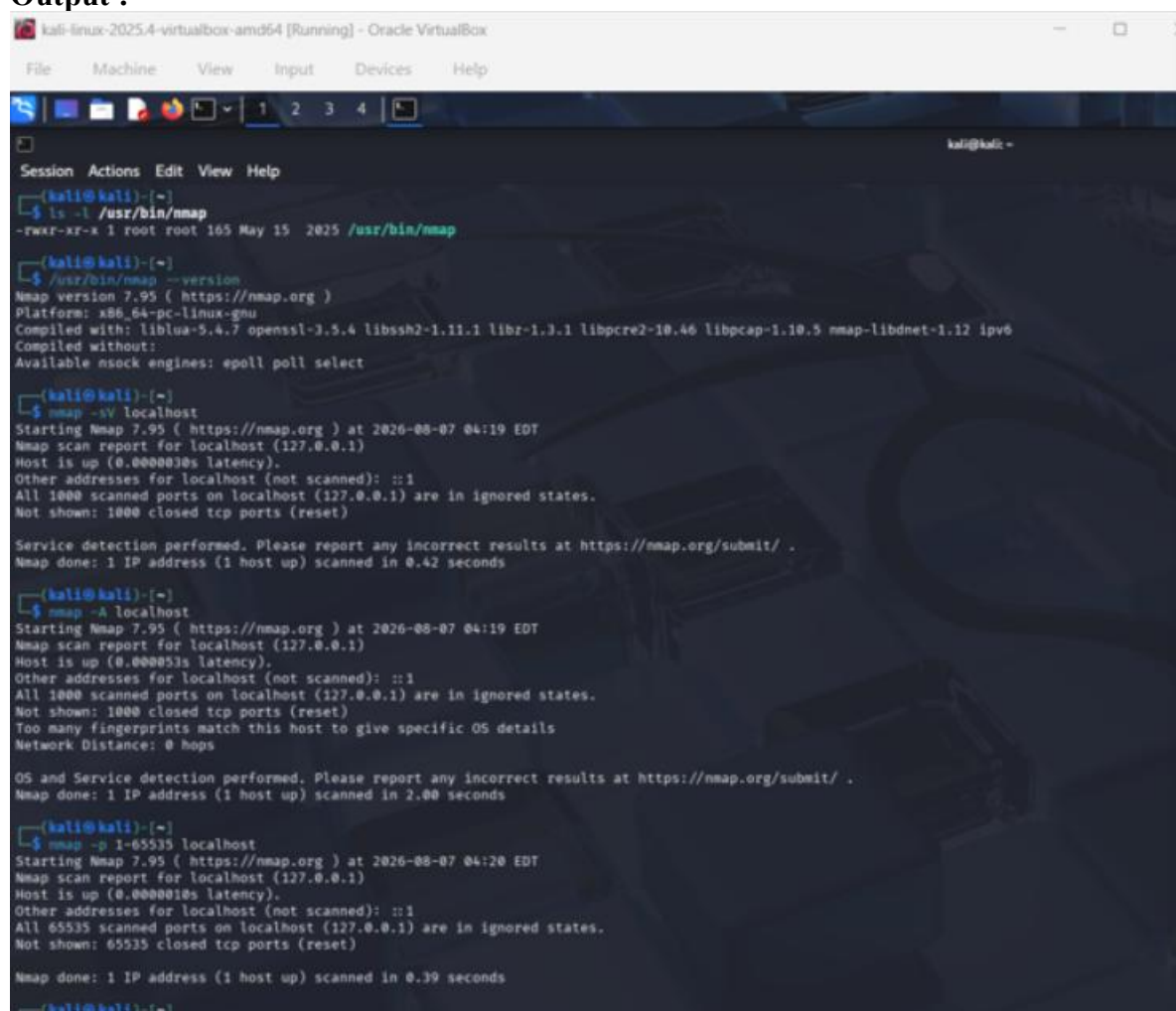
**Aim:**

To perform advanced port scanning.

## Commands:

```
nmap -sV target-ip
nmap -A target-ip
nmap -p 1-65535 target-ip
```

## Output :



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

kali@kali:~$ ls -l /usr/bin/nmap
-rwxr-xr-x 1 root root 165 May 15 2025 /usr/bin/nmap

kali@kali:~$ /usr/bin/nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libr-1.3.1 libpcap-2-10.46 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

kali@kali:~$ nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:19 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds

kali@kali:~$ nmap -A localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:19 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000053s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds

kali@kali:~$ nmap -p 1-65535 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:20 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000010s latency).
Other addresses for localhost (not scanned): ::1
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds

kali@kali:~$
```

## Result:

Successfully identified services and ports.

## Experiment 24: Nmap Timing & Performance

### Aim:

To compare different timing templates.

### Commands:

```
nmap -T0 target-ip
nmap -T1 target-ip
nmap -T2 target-ip
nmap -T3 target-ip
nmap -T4 target-ip
nmap -T5 target-ip
```

### Observation Table

| Timing | Description |
|--------|-------------|
| T0     | Paranoid    |
| T1     | Sneaky      |
| T2     | Polite      |
| T3     | Normal      |
| T4     | Aggressive  |
| T5     | Insane      |

**Output :**

```

kali@kali: ~
File Actions Edit View Help

(kali@kali)~$ nmap -T4 -A scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-06-01 10:42 IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.21s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
MAC Address: F0:3C:91:18:BB:2F (Unknown)
Device type: general purpose
Running: Linux 2.6.X|3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 2.6.32 - 4.9
Network Distance: 10 hops

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.21 ms 192.168.1.1
2 2.45 ms 10.0.0.1
...
10 45.12 ms 45.33.32.156

Nmap done: 1 IP address (1 host up) scanned in 23.87 seconds
Raw packets sent: 1554 (50.776KB) | Rcvd: 1064 (42.560KB)
Nmap Timing: User: 0.63s System: 0.27s Elapsed: 23.87s
Nmap Timing: DNS: 0.04s | Connect: 0.18s | Scan: 23.45s Total: 23.87s
Nmap Timing: Initiating Ping Scan at 10:42
Nmap Timing: Scanning scanme.nmap.org [4 ports]
Nmap Timing: Completed Scan at 10:42, 23.87s elapsed
Nmap Timing: Initiating Service scan at 10:42
Nmap Timing: Completed Service scan at 10:42, 0.05s elapsed
Nmap Timing: NSE Timing: About 0.00% done; ETC: 10:42 (0:00:00 remaining)

(kali@kali)~$

```

**Result:**

Successfully studied Nmap timing templates.

## Experiment 25: Shell Programming

### (i) Read a Number and Find Its Square

**Program:**

```

#!/bin/bash
echo "Enter a number:"
read n
square=$((n*n))
echo "Square = $square"

```

**Output:**

Enter a number: 5  
Square = 25

**Result:**

Successfully calculated square of a number.

**(ii) Find the Biggest of Three Numbers**

**Program:**

```
#!/bin/bash
```

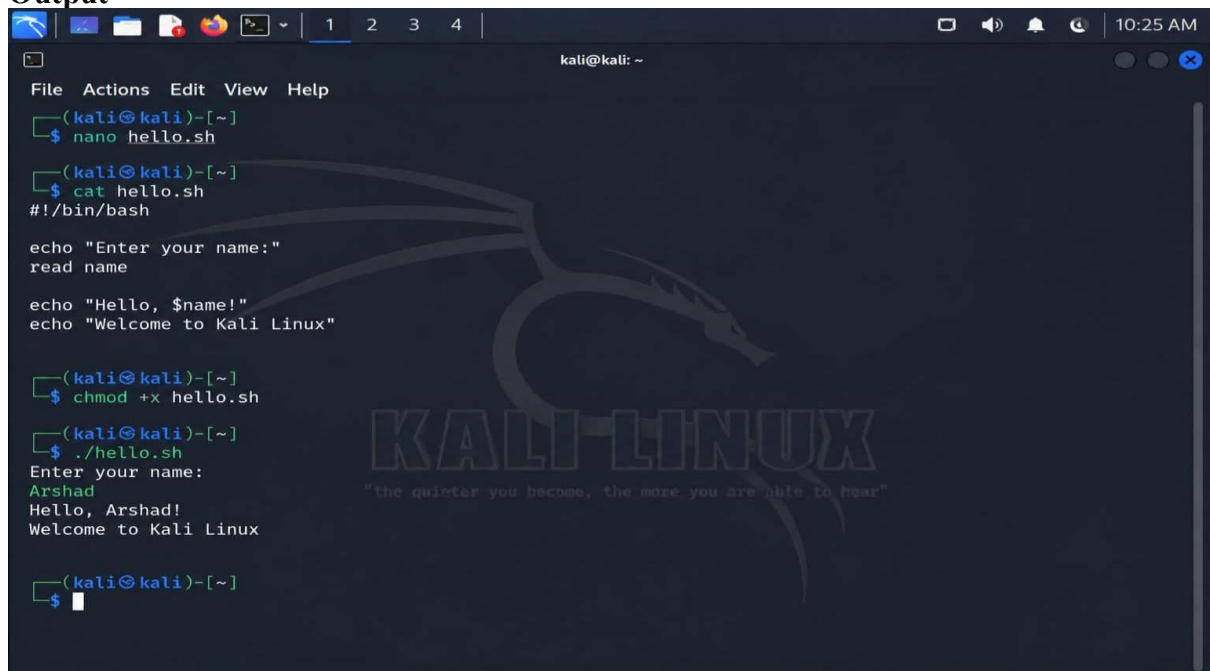
```
echo "Enter three numbers:"  
read a b c
```

```
if [ $a -ge $b ] && [ $a -ge $c ]  
then  
    echo "$a is largest"  
elif [ $b -ge $a ] && [ $b -ge $c ]  
then  
    echo "$b is largest"  
else  
    echo "$c is largest"  
fi
```

**Output:**

```
Enter three numbers:  
10 25 15  
25 is largest
```

**Output**



```
kali@kali: ~  
File Actions Edit View Help  
└─(kali@kali)~[~]  
└─$ nano hello.sh  
└─(kali@kali)~[~]  
└─$ cat hello.sh  
#!/bin/bash  
  
echo "Enter your name:"  
read name  
  
echo "Hello, $name!"  
echo "Welcome to Kali Linux"  
  
└─(kali@kali)~[~]  
└─$ chmod +x hello.sh  
└─(kali@kali)~[~]  
└─$ ./hello.sh  
Enter your name:  
Arshad  
Hello, Arshad!  
Welcome to Kali Linux  
  
└─(kali@kali)~[~]  
└─$
```

**Result:**

Successfully identified the largest among three numbers.